

COLEGIO DE MONTALBAN

Integrated Library Management System (ILMS)

Comprehensive Technical Documentation & Panel Defense Reference

Target: Capstone Panel Defense

Stack: React 18, Node.js, Express, SQLite3

Version: 1.0.0

1. Executive Summary

The **Integrated Library Management System (ILMS)** is an enterprise-grade desktop/web application developed for **Colegio de Montalban (CDM)**. Designed to replace paper logbooks, ILMS automates book circulation, patron management, inventory tracking, overdue fine calculation, and report generation while guaranteeing strict relational data integrity.

2. System Architecture & Tech Stack

Layer	Technologies	Functional Responsibility
Frontend	React 18, TypeScript, Vite 6, Tailwind CSS v4, Lucide Icons, Radix UI	Responsive single-page application (SPA), real-time filtering, visual QR generator, interactive modals.
Backend API	Node.js, Express v5	RESTful service layer providing endpoints for authentication, circulation, inventory, and reporting.
Database	SQLite3 via <code>better-sqlite3</code> driver	Embedded relational database enforcing foreign keys, transactions, and salt-hashed credentials.
Security	PBKDF2 (SHA-512), JWT (HMAC-SHA256)	Password encryption with 16-byte random salts, 24-hour expiration tokens, and role-based authorization.

3. Core Modules & Key Features

- Authentication & Staff Management:** Secure registration with Employee ID, password salt-hashing, role assignment (`Librarian` vs. `Head Librarian`), and mandatory administrative approval for pending staff accounts.
- Analytics Dashboard:** Live counters for Total Books, Active Borrowers, Currently Issued Copies, Overdue Loans, and Pending Reservations alongside daily motivational literacy quotes.
- Book Inventory Management:** Complete CRUD operations, category filters (Literature, History, Science, Math, IT), ISBN uniqueness validation, and automated deletion guards for books with active loans.

- **Circulation & Penalty Engine:** Instant book checkout with default 7-day due date calculation, automated status progression (Active → Overdue), and return condition assessment (Good, Damaged, Lost) with custom fine logging.
- **Patron Directory:** Profile management for students, including Course (BSIT, BSCE, BSEd), Year Level, contact information, and protection against deleting patrons with active library holdings.
- **Official Report Exporting:** Instant generation of styled, official Colegio de Montalban **Transaction Reports** and **Book Inventory Audits** formatted for printing/PDF export.

4. Database Schema

```
LIBRARIANS (id INTEGER PK, first_name, last_name, email UNIQUE, employee_id UNIQUE, role, usernam
BOOKS      (id TEXT PK, title, author, isbn UNIQUE, category, cover, abstract, available INT, tot
STUDENTS   (id TEXT PK, name, email UNIQUE, phone, course, year_level, status)
TRANSACTIONS (id TEXT PK, book_id FK, book_title, student_name, student_id, librarian_name, borrow_
RESERVATIONS (id TEXT PK, book_id FK, book_title, student_name, student_id, reservation_date, picku
```

5. Defense Panel FAQ Cheatsheet

Q1: Why SQLite instead of MySQL or PostgreSQL?

SQLite with `better-sqlite3` delivers zero-configuration, lightning-fast synchronous queries directly from a local database file (`library.db`). This eliminates database server management while guaranteeing local data persistence for school library workstations.

Q2: How do you prevent inventory count errors during concurrent borrowing?

All checkout and return routines are wrapped in `db.transaction()` calls. If any step fails or an item is out of stock, SQLite rolls back all changes automatically, preserving exact stock numbers.

Q3: How is password authentication secured against security attacks?

Passwords are multi-iterated (1,000 rounds of PBKDF2 with SHA-512) paired with a 16-byte random salt per user. Timing-safe comparisons are used during login to protect against timing attacks.

Q4: What happens when a book is reported lost by a student?

The return flow records the condition as `lost` and logs the penalty fee. The transaction marks as completed, but available shelf stock is **not incremented**, maintaining accurate real-world counts.